

USACE Savannah District: Mathematical & Regulatory Framework

Quantitative Credit Calculations, Developer Debit Formulas, & RIBITS Ledger Mechanics

This framework establishes the exact mathematical models, credit/debit worksheets, and administrative processes utilized by the **U.S. Army Corps of Engineers (USACE) Savannah District** to govern stream compensatory mitigation in Georgia. It serves as Hunter's definitive guide to calculating credit yields and understanding how developers are mathematically assessed for their environmental impacts.

1. How the U.S. Army Corps of Engineers (USACE) Operates

The USACE is organized geographically into divisions and districts. Georgia falls under the **South Atlantic Division (SAD)**, and the **Savannah District** holds regulatory jurisdiction over all wetlands and streams in the state.

The Regulatory Branch of the Savannah District operates under the **2008 Compensatory Mitigation Rule** (33 CFR Parts 325 and 332). The District is co-chaired by the **Interagency Review Team (IRT)**, which includes:

- U.S. Army Corps of Engineers (USACE) — *Savannah District (Chair)*
- U.S. Environmental Protection Agency (EPA) — *Region 4*
- U.S. Fish and Wildlife Service (USFWS)
- National Marine Fisheries Service (NMFS)
- Georgia Department of Natural Resources (DNR) — *Environmental Protection Division (EPD)*

All commercial mitigation banks must be approved by the IRT through a consensus process, whereas Permittee-Responsible Mitigation (PRM) projects (such as Anderson Creek) are approved directly by the USACE Project Manager in coordination with EPD stream buffer variance regulators.

2. The Developer's Debit Formula (Calculating Required Credits)

When a developer (such as **Fuqua Development** or **Sefried Development**) impacts a stream, the USACE Savannah District assesses the impact using the **Adverse Impact Worksheet**. This worksheet calculates the total number of **debits** (credits the developer must purchase from a bank to achieve compliance).

The Mathematical Model for Adverse Impact (Debits)

$$\text{Total Debits Required (D)} = \text{Stream Length Impacted (LF)} \times \sum (\text{Impact Factors}) \times R$$

Where:

- **Stream Length Impacted (LF)**: The linear footage of stream channel physically disturbed, piped, filled, or culverted.

- **$\sum (\text{Impact Factors})$** : The sum of values assigned to four distinct qualitative and quantitative parameters:
 1. **Stream Type Factor (F_{type})**: Measures the flow regime of the impacted stream.
 - Perennial (flowing year-round): **0.9**
 - Intermittent (flowing seasonally): **0.4**
 - Ephemeral (flowing only after rain): **0.1**
 2. **Impact Type Factor (F_{impact})**: Measures the severity of the structural impact.
 - Piping/Culverting (complete stream loss): **2.0**
 - Channelization/Relocation (concrete or geogrid lining): **1.5**
 - Clearing/Grubbing (vegetation removal only): **0.5**
 3. **Duration of Impact Factor (F_{duration})**:
 - Permanent (structure remains in perpetuity): **1.5**
 - Temporary (< 1 year for utility crossing): **0.2**
 4. **Watershed Priority Factor (F_{priority})**: High-priority basins identified in the district's River Basin Management Plans.
 - Primary Priority Watershed (e.g., active trout streams): **0.4**
 - Secondary Priority Watershed: **0.1**
 - **R (Regional Multiplier)**: Savannah District regional scaling factor (typically set to **1.0**).
-

■ Practical Example: Developer Debit Calculation

Consider a project proposed by **Fuqua Development** that requires piping **1,200 linear feet** of a perennial trout stream in a high-priority watershed (e.g., Coosa River basin) to build a retail shopping center:

1. Parameters:

- Stream Length = 1,200 LF
- Stream Type (F_{type}) = Perennial $\rightarrow$ **0.9**
- Impact Type (F_{impact}) = Piping $\rightarrow$ **2.0**
- Duration (F_{duration}) = Permanent $\rightarrow$ **1.5**
- Priority (F_{priority}) = Primary $\rightarrow$ **0.4**

2. Sum of Factors:

$$\sum (\text{Factors}) = 0.9 + 2.0 + 1.5 + 0.4 = 4.8$$

3. Total Debits Required:

$$\text{Debits} = 1,200 \text{ LF} \times 4.8 = \mathbf{5,760 \text{ Credits}}$$

4. Financial Cost to Developer:

If stream credits in the basin trade at **\$110/credit**, Fuqua Development must purchase 5,760 credits at a total cost of **\$633,600** from a mitigation bank to secure their permit.

3. The Bank's Credit Formula (Calculating Credit Generation)

To determine how many **credits** Hunter's stream restoration projects generate, the USACE Savannah District applies the **Mitigation Credit Worksheet**. This worksheet calculates the total ecological lift based on instream restructuring and riparian buffer enhancements.

The Mathematical Model for Stream Mitigation Credits (Credits)

$\text{Credits Generated (C)} = \text{Restored Length (LF)} \times \sum (\text{Mitigation Factors}) \times W$
Where:

- **Restored Length (LF)**: The linear footage of stream channel physically restored.
- **Sum (Mitigation Factors)**: The sum of values assigned to six ecological lifting parameters:
 1. **Net Benefit Factor (M_{benefit})**: The scale of geomorphic and biological lift.
 - *Priority 1 Restoration* (Rosgen Natural Channel Design, reconnecting stream to historic floodplain footing): **4.0**
 - *Priority 2 Restoration* (Creating a new, stable floodplain bench at current elevation): **3.0**
 - *Priority 3 / Bioengineering* (Bank grading and soft armoring with live stakes/root wads): **1.5**
 2. **Riparian Buffer Factor (M_{buffer})**: Evaluated per bank (total of Left + Right banks).
 - 100-foot buffer planted with native woody species: **1.2 per bank** (Total: **2.4**)
 - 50-foot buffer planted with native woody species: **0.6 per bank** (Total: **1.2**)
 3. **Monitoring Factor ($M_{\text{monitoring}}$)**:
 - 7-Year Trout Stream Commitment (including thermal and EPT macroinvertebrate scoring): **0.4**
 - 5-Year Standard Commitment: **0.2**
 4. **Site Protection Factor ($M_{\text{protection}}$)**:
 - Perpetual Restrictive Covenant/Easement held by third-party Land Trust: **1.0**
 5. **Stream Flow Factor (M_{flow})**:
 - Perennial: **0.4**
 - Intermittent: **0.2**
 6. **Watershed Priority Factor ($M_{\text{watershed}}$)**:
 - Primary Priority Watershed (e.g., North Georgia trout waters): **0.4**
- **W (Watershed Multiplier)**: Adjusts credit yield based on structural location (typically set to **1.0**).

■ Practical Example: Anderson Creek (ACR) Credit Yield

Using Hunter's geomorphic design for the **30-acre Anderson Creek lot** across **1,500 linear feet** of degraded perennial stream:

1. **Parameters:**
 - Stream Length = 1,500 LF
 - Net Benefit (M_{benefit}) = Priority 1 Geomorphic Restoration $\rightarrow$ **4.0**
 - Riparian Buffer (M_{buffer}) = 100-foot buffer on both banks $\rightarrow 1.2 \times 2 = \mathbf{2.4}$
 - Monitoring ($M_{\text{monitoring}}$) = 7-Year Trout SOP $\rightarrow$ **0.4**
 - Site Protection ($M_{\text{protection}}$) = Perpetual Conservation Easement $\rightarrow$ **1.0**

- Stream Flow (M_{flow}) = Perennial $\rightarrow$ **0.4**
- Watershed Priority ($M_{\text{watershed}}$) = Primary Trout Watershed $\rightarrow$ **0.4**

2. **Sum of Mitigation Factors:**

$$\sum (\text{Factors}) = 4.0 + 2.4 + 0.4 + 1.0 + 0.4 + 0.4 = \mathbf{8.6}$$

3. **Total Credits Generated:**

$$\text{Credits} = 1,500 \text{ LF} \times 8.6 = \mathbf{12,900 \text{ Credits}}$$

4. **Gross Asset Value:**

At a premium coldwater trout basin credit price of **\$180/credit** (due to extreme scarcity in North Georgia), the total credit asset generated from the Anderson Creek lot is **\$2,322,000!**

4. RIBITS and Ledger Mechanics

All stream and wetland mitigation banks are tracked in a public federal database called **RIBITS** (Regulatory In-lieu fee and Bank Information Tracking System), maintained by the USACE.

RIBITS Ledger Operations

- **Credit Ledger:** Every approved bank has a credit ledger in RIBITS. Credits are added to the ledger upon reaching specific regulatory milestones (credit releases) and are deducted when sold to developers.
- **Credit Releases:** Credits are not released all at once. The Savannah District operates under a standard **Credit Release Schedule**:
 1. **Initial Release (15%):** Released immediately upon signing the Mitigation Banking Instrument (MBI), recording the perpetual Conservation Easement, and funding the Financial Assurances. (At Anderson Creek, this equals **1,935 credits** or **\$348,300** in immediate tradable value).
 2. **Construction Release (25%):** Released upon completion of all instream geomorphic earthwork and USACE approval of the as-built survey.
 3. **Annual Success Releases (60%):** Released in equal increments of **10%** over the subsequent 6 years, contingent on satisfying the geomorphic stability, vegetation survival, and water temperature success criteria.

Financial Assurances

To secure credit releases, the Bank Sponsor must establish a **Financial Assurance Account** (typically in the form of a **Performance Bond** or **Letter of Credit** held in escrow).

- **Purpose:** Protects the USACE in case the bank sponsor goes bankrupt or fails to maintain the stream channel during the 7-year monitoring phase.
- **Funding:** The bond is typically funded at a rate of **\$15 to \$25 per linear foot** of restored stream. For Anderson Creek, this represents a capital bond of **\$30,000 to \$50,000**, which is fully returned upon completing the 7-year monitoring period successfully.